Université Ibnou Zohr
Faculté des Sciences - Agadir
Département de Chimie
Année Universitaire 2023-2024

PC – PHYSIQUE (S_2)
Module (M_{127}) : Chimie Organique Générale
Corrigé - TD : Série N° 2

Nomenclature des composés organiques

Exercice I

Nommer les composés suivant selon la nomenclature officielle IUPAC :

1) 3,5-diméthyloctane ;

2) 3-éthyl-4-isopropyl-2,2,5-triméthylheptane ;

3) 4-chloro-5-éthyl-3,6-diméthyloctane ;

4) 2,3-diméthylpentane ;

5) 4,5-diéthyl-3-méthyloctane ;

6) 6-fluoro-2,3,7-triméthylnonane ;

7) 4-méthyl-3-propylhex-2-éne ;

8) 4-éthylhex-1-én-5-yne ;

9) 5-bromohex-4-én-1-yne ;

10) 1-éthyl-3-isopropyl-5-méthylcyclohexane ;

11) 3,3-diméthyl-5-nitrocyclopent-1-éne ;

12) 1-isopropyl-4-méthylcyclohexa-1,3-diène ;

13) 5-méthylcyclopent-2-énol ;

14) 5-chloro-3-éthyl-3-méthylcyclohexanone ;

15) 2,4-diméthylpentan-3-amine (1-isopropyl-2-méthylpropylamine);

16) N-éthyl-N-méthylpent-4-én-2-amine (N-éthyl-N-diméthyl-1-méthylbut-3-énylamine);

17) 3-hydroxybutanenitrile ;

18) 4-amino-3-méthylpentan-2-one ;

19) 3-éthyl-5-methoxyhexanamide ;

20) méthyl 2-propylbut-3-énoate ;

21) acide 3-formylpentanoïque ;

22) chlorure de 3-methoxypentanoyle ;

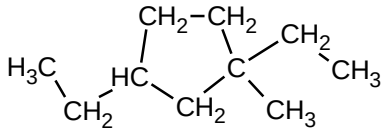
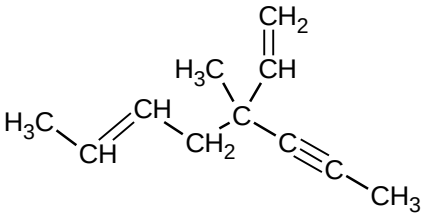
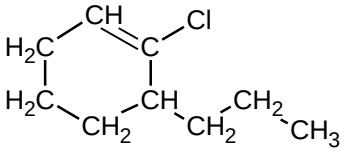
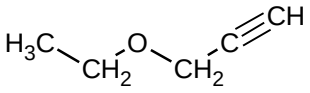
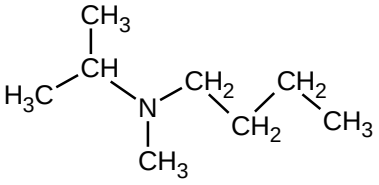
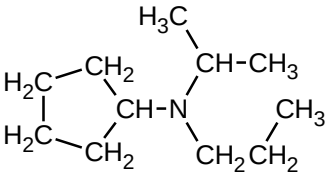
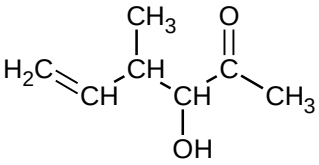
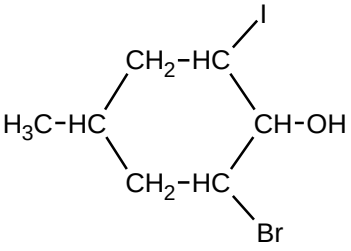
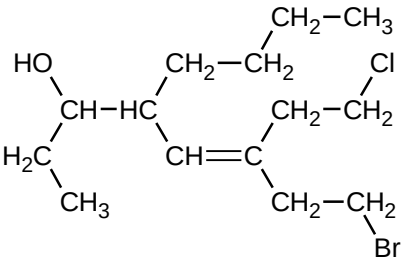
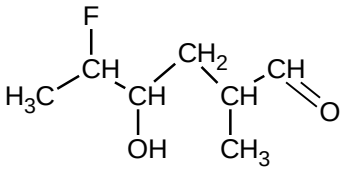
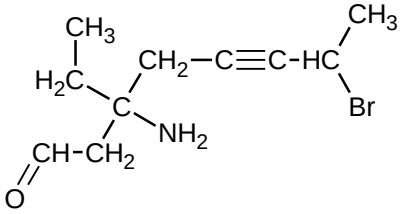
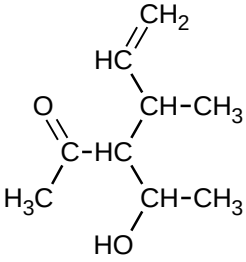
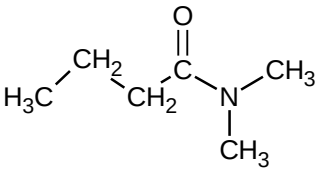
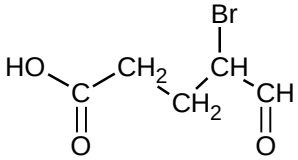
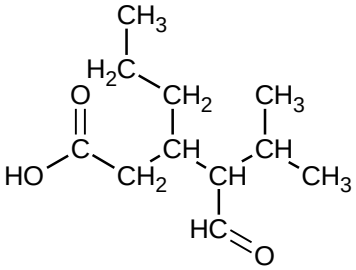
23) acide 4-cyanopent-2-énedioïque ;

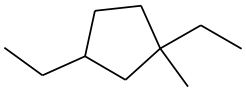
24) acide 3-éthylhexanoïque ;

25) 5-éthyl-4,7-dioxooctanal.

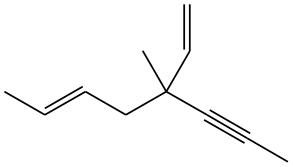
Exercice II

Donner la formule semi-développée de ces composés ci-après :

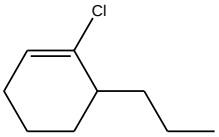
 **1-Méthyl-1,3-diéthylcyclopentane**	 **5-Méthyl-5-vinyloct-2-én-6-yne**	 **2-Chloro-3-propylcyclohexène**
 **3-Ethoxyprop-1-yne**	 **N-Isopropyl N-méthylbutan-1- amine (N-Isopropyl N-méthylbutylamine)**	 **N-Isopropyl-N-propylcyclopentylamine (N-isopropyl-N-propylcyclopentanamine)**
 **3-Hydroxy-4-méthylhex-5-èn-2-one**	 **2-Bromo-6-iodo-4-méthylcyclohexanol**	 **8-Bromo 4-butyl-6-(2-chloroéthyl)oct-5-èn-3-ol**
 **5-Fluoro-4-hydroxy-2-méthylhexanal**	 **3-Amino-7-bromo-3-éthyloct-5-ynal**	 **3-(1-Hydroxyéthyl)-4-méthylhex-5-èn-2-one**
 **N,N-Diméthylbutanamide**	 **Acide 4-bromo-5-oxopentanoïque**	 **Acide 4-formyl-5-méthyl-3-propylhexanoïque**



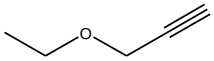
1) 1,3-Diéthyl-1-méthylcyclopentane



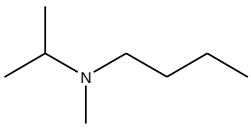
2) 5-Méthyl-5-vinyloct-2-én-6-yne



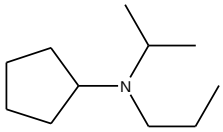
3) 2-Chloro-3-propylcyclohexène



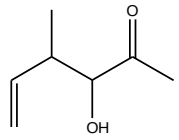
4) 3-Ethoxyprop-1-yne



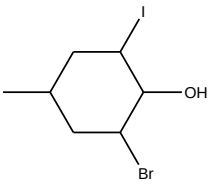
5) N-Isopropyl N-méthylbutan-1- amine
(N-Isopropyl N-méthylbutylamine)



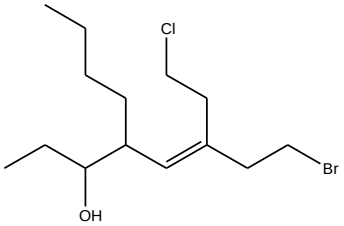
6) N-Isopropyl-N-propylcyclopentylamine
(N-isopropyl-N-propylcyclopentanamine)



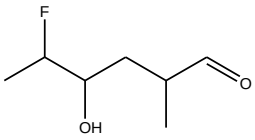
7) 3-Hydroxy-4-méthylhex-5-èn-2-one



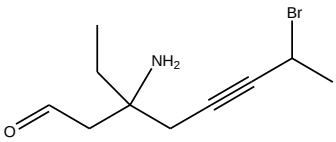
8) 2-Bromo-6-iodo-4-méthylcyclohexanol



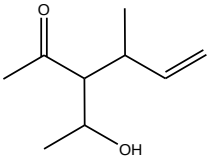
9) 8-Bromo 4-butyl-6-(2-chloroéthyl)oct-5-èn-3-ol



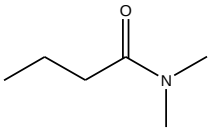
10) 5-Fluoro-4-hydroxy-2-méthylhexanal



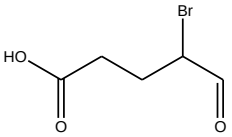
11) 3-Amino-7-bromo-3-éthyloct-5-ynal



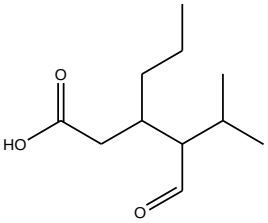
12) 3-(1-Hydroxyéthyl)-4-méthylhex-5-èn-2-one



13) N,N-Diméthylbutanamide



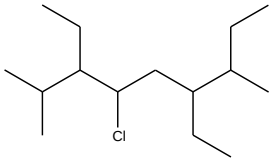
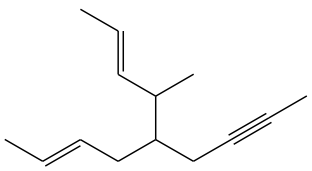
14) Acide 4-bromo-5-oxopentanoïque

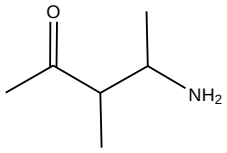
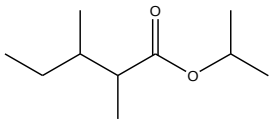
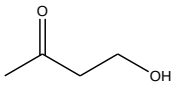
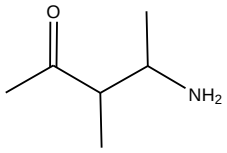
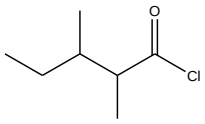
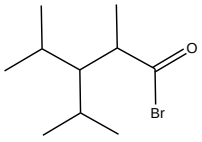
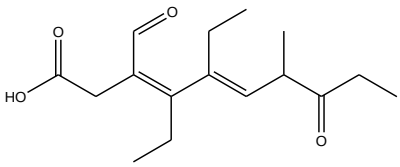


15) Acide 4-formyl-5-méthyl-3-propylhexanoïque

Exercice III

Ecrire en formule simplifiée (FS) et vérifier si les noms suivants correspondent à la nomenclature systématique :

1) 6-Chloro-4-éthyl-3-méthyl-7-isopropylnonane (Faux) **4-chloro-6-éthyl-3-isopropyl-7-méthylnonane**	
2) 5-(But-2-ényl)-4-méthylnon-2-én-7-yne (Faux) **5-(but-2-ynyl)-4-méthylnona-2,7-diène**	

3) 3-Méthyl-2-oxo-pentan-4-amine (Faux) **4-amino-3-méthylpentan-2-one**	
4) 2,3-Diméthylpentanoate de 1-méthyléthyle (correcte) **2,3-Diméthylpentanoate d'isopropyle**	
5) 3-Oxobutan-1-ol (Faux) **4-hydroxybutan-2-one**	
6) 4-Oxo-3-méthylpentan-2-amine (Faux) **4-amino-3-méthylpentan-2-one**	
6) Chlorure de 2,3-diméthylpentanoyle (correcte)	
7) Bromure de 2,4-diméthyl-3-isopropylpentanoyle (Faux) **Bromure de 3-isopropyl-2,4-diméthylpentanoyle**	
9) Acide de 4,5-diéthyl-3-formyl-7-méthyl-8-oxodéca-3,5-diénoïque (correcte)	
10) 4-oxohept-6-amino-2-énoate de propyle (Faux) **6-amino-4-oxohept-2-enoate de propyle**	